

“THE HEAT WHICH DISAPPEARS IN THE CONVERSION OF WATER INTO VAPOR, IS NOT LOST.”

Joseph Black, Scottish chemist, from his *Lectures on Elements of Chemistry*, 1960



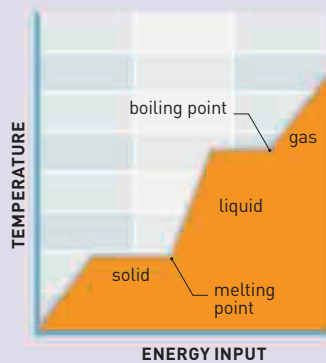
Joseph Black became famous for his groundbreaking lectures on the study of heat at Glasgow University, Scotland.

ONE OF THE MILESTONES on the road to the Industrial Revolution was the establishment in 1761 of the **Soho Manufactory** in Birmingham, England. The brainchild of entrepreneur Matthew Boulton, the Soho factory pioneered the **assembly line**, with mass-production of cheap items, such as buttons, buckles, and boxes for the general public. It was here that British engineer **James Watt's steam engines** (see 1765–66) were built a few years later.

An early patron of James Watt, Scottish chemist **Joseph Black** (1728–99) discovered one of the properties of heat. He found that much more heat is needed to melt ice than to warm ice-cold water, just as it takes extra heat to evaporate water. The discovery

LATENT HEAT

Substances absorb or release heat when their physical state undergoes change. For a solid to melt to a liquid, it must absorb heat energy; and when a liquid freezes to a solid, it loses heat. This energy is called latent heat. Much more heat energy is needed to change solids to liquids and liquids to gases than to simply raise the temperature of a substance.



of this **latent heat**, highlighting the difference between heat and temperature, was fundamental to the modern understanding of energy. It also helped Watt turn the steam engine from an inefficient contraption to the powerhouse of the Industrial Age.

In Sweden, chemist Johan **Gottschalk Wallerius** (1709–85) showed how science could be applied to farming as well as to industry. His pioneering work on **agricultural chemistry**, *Agriculturae Fundamenta Chemica* (*The Natural and Chemical Elements of Agriculture*), discussed the chemical components most conducive to plant growth.

Meanwhile, another Swede, Johan Carl Wilcke (1732–96) designed a **dissectible**

condenser—a device to generate static electricity.

In Switzerland, mathematician **Johann Heinrich Lambert** (1728–77) published a treatise in which he put forward his version of the **nebular hypothesis**, the theory that the Solar System developed from a cloud of interstellar dust. English astronomer Thomas Wright and German philosopher Immanuel Kant independently proposed a similar theory. Lambert also suggested rightly that the Sun and nearby stars travel together as a group through the Milky Way.

100+
STARBURST
GALAXY

1
An ordinary galaxy

Stars made by galaxies per year
The Milky Way, the only known galaxy in the 1760s, makes one new star a year, compared to hundreds in the “starburst” galaxies now recognized.



Edward Stone's discovery of the medicinal properties of willow bark was a great breakthrough in palliative medicine.

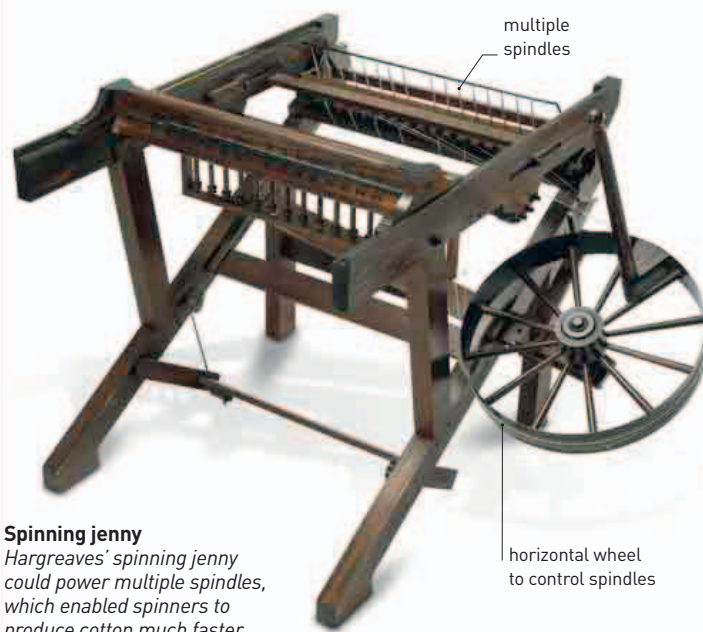
IN 1763, BRITISH CLERGYMAN EDWARD STONE (1702–68) discovered the **medicinal properties of willow bark**.

Stone found, after careful testing, that willow bark dramatically reduced ague, a fever with symptoms similar to malaria. It was later found that the active ingredient in willow bark is salicylic acid, which forms the basis of aspirin (see 1897).

One of the most influential inventions of 1764 was English carpenter and weaver **James Hargreaves' (1721–78) spinning jenny**, which could spin cotton eight times faster than a manual

worker. The spinning jenny was hand-driven and required some skill to operate, but it was a key step toward automatic, powered machines that could turn out cloth in vast quantities.

In 1764, Italian-born French mathematician **Joseph Louis Lagrange** (1736–1813) showed why the **Moon oscillates, or librates**, continually. He also explained why the same lunar face is always turned toward Earth. This explanation later became the basis of his equations of motion, which provide a simple way of calculating the movement within a system.



Spinning jenny
Hargreaves' spinning jenny could power multiple spindles, which enabled spinners to produce cotton much faster.

1761 Johan Gottschalk Wallerius pioneers agricultural chemistry

June 6, 1761
Transit of Venus is widely observed

1761 Joseph Black discovers latent heat

1762 Johan Carl Wilcke designs electrostatic generator

1764 James Hargreaves invents the spinning jenny

1764 Joseph Louis Lagrange shows the libration of the Moon